

LULEA KALLAX, Sweden

WMO: 021860

Lat: 65.543N

Lon: 22.119E

Elev: 65

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.4	-9.6	-20.3	1.8	-15.3	-14.1	2.6	-9.5	26.0	32.1	23.1	31.0	3.0	320	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	74.9	62.9	71.9	61.6	69.4	60.4	65.9	72.0	63.9	69.1	61.9	67.0	8.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	88.9	68.8	61.6	82.0	66.7	59.4	75.9	64.5	30.7	72.0	29.2	69.1	27.8	67.2	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.5	16.1	-21.4	80.8	7.0	4.5	-26.4	84.1	-30.5	86.7	-34.4	89.2	-39.5	92.4		
(5)			-21.7	67.6	6.9	2.4	-26.7	69.3	-30.7	70.8	-34.6	72.1	-39.6	73.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	16.0	15.2	23.5	33.6	44.4	55.0	61.3	58.0	49.1	37.5	27.0	20.2		
	DBStd	18.18	12.80	13.35	9.41	6.07	6.71	5.36	4.91	5.25	5.93	7.31	9.76	12.34		
	HDD50	5647	1053	975	821	491	199	14	0	4	87	389	691	923		
	HDD65	10299	1518	1395	1286	941	638	302	136	224	477	853	1141	1388		
	CDD50	855	0	0	0	0	27	164	350	253	60	1	0	0		
	CDD65	32	0	0	0	0	0	2	22	8	0	0	0	0		
	CDH74	135	0	0	0	0	3	16	99	17	0	0	0	0		
	CDH80	14	0	0	0	0	0	2	11	1	0	0	0	0		
	WSAvg	7.1	6.8	6.9	7.5	6.8	7.4	7.7	7.2	7.0	7.4	7.3	7.0	6.8		
	PrecAvg	20.9	1.5	1.3	1.3	1.2	1.2	1.6	2.0	2.6	2.2	2.0	2.2	1.8		
(15) Precipitation	PrecMax	26.8	3.5	2.4	3.1	2.6	2.6	3.8	5.9	5.5	4.1	4.4	3.9	5.9		
	PrecMin	15.0	0.2	0.2	0.1	0.4	0.2	0.2	0.4	0.7	0.6	0.5	0.7	0.5		
	PrecStd	3.0	0.7	0.6	0.9	0.7	0.6	1.0	1.2	1.3	0.8	1.0	0.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	44.7	48.1	55.5	71.1	75.9	81.0	76.0	66.2	55.2	47.8	43.2		
		MCWB	39.1	39.8	40.9	45.5	57.7	59.3	65.7	64.2	56.5	47.6	43.4	39.7		
	2%	DB	37.6	39.9	43.7	50.5	64.8	71.2	76.7	72.4	62.7	51.7	43.9	39.5		
		MCWB	35.0	35.9	37.6	42.1	53.2	58.3	64.6	62.9	55.5	48.7	42.3	37.1		
	5%	DB	34.4	35.4	40.1	46.8	60.4	67.8	73.5	69.7	60.4	49.8	41.3	37.2		
		MCWB	33.2	33.1	35.1	39.8	50.3	56.7	62.6	61.7	54.8	47.8	40.2	35.7		
	10%	DB	32.4	32.9	36.1	43.3	56.0	64.7	70.6	67.3	58.3	47.9	39.3	35.4		
		MCWB	31.6	31.8	32.7	37.7	47.5	55.2	61.6	60.4	54.2	45.8	38.0	34.3		
	0.4%	WB	39.3	40.0	41.3	46.5	58.4	64.2	69.0	68.2	59.7	51.5	44.7	40.1		
		MCDB	42.4	44.3	47.0	54.6	68.3	70.1	75.8	73.2	62.9	52.6	45.7	42.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	36.2	38.1	43.0	54.7	60.6	66.8	65.3	57.7	49.7	42.7	37.6		
		MCDB	37.2	39.2	42.9	49.1	63.3	66.5	73.7	69.7	60.6	50.6	43.7	38.9		
	5%	WB	33.5	33.4	35.3	40.6	51.1	58.7	65.1	63.2	56.4	47.8	40.4	35.9		
		MCDB	34.3	34.6	38.8	45.8	58.5	65.0	71.1	67.6	59.0	49.4	41.3	36.8		
	10%	WB	31.9	32.0	33.3	38.2	48.4	56.7	63.2	61.1	55.1	45.7	38.1	34.2		
		MCDB	32.5	33.0	35.4	42.6	54.9	63.1	68.6	65.6	57.6	47.4	39.1	35.1		
	5% DB	MDBR	13.3	14.3	15.1	13.8	14.6	13.1	12.8	12.6	12.3	10.2	9.9	11.8		
		MCDBR	13.0	16.5	17.1	18.9	20.6	17.7	17.1	15.4	14.3	11.3	9.5	10.4		
		MCWBR	11.9	13.8	12.5	12.2	11.5	8.9	8.2	8.5	9.1	8.8	8.8	9.5		
		MCWBR	11.4	13.3	15.5	18.2	19.5	15.2	14.1	13.0	11.2	9.3	8.6	9.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.7	11.6	11.9	12.2	11.8	9.0	7.9	8.3	8.1	8.0	8.3	8.9		
	taub		N/A	0.272	0.301	0.335	0.343	0.347	0.362	0.347	0.319	0.298	0.340	0.349		
		taud	N/A	2.135	2.208	2.282	2.442	2.490	2.468	2.500	2.554	2.382	2.610	2.633		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		N/A	189	242	263	274	275	266	259	243	185	84	30		
		Edh at Noon	N/A	18	27	32	30	30	30	26	20	14	6	3		
	RadAvg		30	208	699	1272	1602	1786	1647	1211	668	247	49	7		
(44) All-Sky Solar Radiation	RadStd		3	36	62	73	147	139	147	109	90	45	8	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-257	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page